Banking Loan Risk Analysis Application

Technical Evaluation, Test Benchmark & Insights Report

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1. Evaluation Methodology & Verification Scope

The solution was evaluated across data quality robustness, reconciliation accuracy, risk engine precision, API response latency, and database query performance on MySQL.

2. Automated Test Suite Results

A 15-test automated test suite was executed via Pytest:

- **ETL Pipeline & Batch Moving:** Verified raw extraction, categorical decoding, upsert semantics, and automatic file movement from data/raw/ to data/processed/ (Passed).
- **Business Rules & Risk Engine:** Verified Low, Medium, and High risk band classifications across edge cases, as well as high-exposure and chronic delinquency review triggers (Passed).
- **Data Quality & Reconciliation:** Validated duplicate checks, range checks, MD5 checksum computation, and record reconciliation audits (Passed).
- **SQL & Relational DB Tests:** Validated ACID CRUD operations, foreign key cascades, and pre-aggregated KPI tables on MySQL (Passed).
- **FastAPI Endpoints:** Validated JWT RBAC, paginated querying, risk evaluation, and KPI aggregations (Passed).

3. Technical Performance Benchmarks

- **Ingestion Throughput:** 3,000 records processed in 3.19s with 100% cryptographic reconciliation.
- **File Lifecycle Automation:** Processed raw batches moved cleanly to data/processed/ with timestamped audit names.
- **API Latency:** Under 25ms for paginated customer queries; under 15ms for risk evaluation.
- **Data Quality Resilience:** 8 distinct rule suites executed per batch with zero unhandled exceptions.

4. Key Portfolio Insights Discovered

- **Education Breakdown:** High School segment exhibited the highest default rate (26.17%) and highest high-risk concentration (48.88%), compared to Graduate School clients (19.23% default rate).
- **Age Trends:** Default rate peaks in the 60+ age group (28.57%) and 50-59 group (26.96%), while lowest in the 30-39 bracket (20.96%).
- **Utilization Indicator:** Customers with credit utilization above 85% accompanied by payment delays exhibited default rates exceeding 68%, confirming the validity of the High Risk classification rule.